Quality Control Final Report

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1. Introduction

Morden manufacturing systems are huge issues. Lots of sensors generate high-frequency, multivariate time-series data across many machines and production lines. From an operations perspective, the ability to detect anomalies early in such data is crucial. However, modern sensor data differ significantly from these traditional assumptions. Production equipment is typically monitored by dozens of correlated variables, and the data sets are long, noisy, and often nonstationary. Directly applying conventional SPC to each variable separately ignores cross-variable relationships and may fail to capture subtle multivariate patterns that indicate early-stage faults. This gap motivates the use of machine learning methods that can learn compact representations from high-dimensional data and provide richer anomaly signals. To address these challenges, this work combines autoencoder with Gaussian noise, robust covariance estimation and constructing the two control charts. We train a denoising autoencoder on normal data to compress multivariate time windows into a low-dimensional latent space. In the learned latent space, we fit a robust Gaussian model using the Minimum Covariance Determinant (MCD) estimator and compute a latent-space Mahalanobis distance as a primary anomaly statistic. In parallel, we use the autoencoder's reconstruction error as complementary statistics. Through these steps, this work hopes that combining an AE-based latent representation, robust MCD modeling, and draw the control charts can provide a more stable and interpretable anomaly

detection system and also construct PCA-MSPC control charts for this dataset.

2. Research Problem

2.1 Related work

Autoencoders (AEs) are widely used for anomaly detection in time-series sensor data. A common pipeline first trains an AE on windows extracted from normal data and then applies the trained model to unseen windows, where the reconstruction error serves as an anomaly indicator: windows that the model cannot reconstruct well are more likely to be abnormal. Building on this line of work, we train an autoencoder with Gaussian noise on normal windows and use both the latent representation and the reconstruction error as anomaly indicators, rather than relying on a single scalar score.

For multivariate monitoring, Mahalanobis distance (MD) is a classical way to summarize how far a point is from the "normal cloud" in a high-dimensional space. Given a mean vector μ and covariance matrix Σ, the distance:

$$d(x) = \sqrt{(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

provides a single statistic that combines all variables and their correlations. Traditional Hotelling T^2control charts are essentially built on this idea. To improve robustness, Minimum Covariance Determinant (MCD) estimators have been proposed. MCD searches for a subset of data whose covariance determinant is minimal and uses this subset to estimate a clean mean and covariance. In contrast to those PCA-based approaches, this project applies MCD directly in the latent space of a deep autoencoder, which is designed to capture non-linear structure in the data.

In this project, to turn deep-learning–based scores into interpretable monitoring charts.

The latent-space MD and reconstruction error are treated as chart statistics, and Shewhart / EWMA limits are set by empirical quantiles so that the in-control false alarm rate is approximately controlled.

This project combines these components into a hybrid framework: A AE with Gaussian Noise provides a low-dimensional latent representation of high-dimensional sensor windows. MCD-based Mahalanobis distance in latent space and the reconstruction error are both used as anomaly statistics. These statistics are monitored by Shewhart and EWMA control charts with quantile-based upper control limits, and their binary decisions are fused using different logical rules (OR / AND / VOTE).The impact of the quantile level q, smoothing parameter λ, and fusion logic on precision, recall, F1, and point-adjusted metrics is systematically evaluated on the SMD dataset.

2.2 Dataset and preprocessing

This project uses Server Machine Dataset, which is a public benchmark for anomaly detection in data center servers. The datasets include 28 machines, and each machine has 38 sensors. SMD provides an official split into a training sequence and a test sequence. We split the official test sequence into validation set (30% of the test data) and a final test set (about 70%). In this work we focus on machine 1-1 and select 10 sensors as our datasets to do. The time series becomes a multivariate sequence with 10 variables, so each time index t is represented by a 10-dimensional vector. The split is performed at the level of anomalous segments rather than individual points: consecutive anomalous points are grouped into segments, and some segments are assigned to the validation set while the others remain in the final set, with normal points filled in around them. This “stratified by segments” strategy keeps the anomaly ratio of validation and final test reasonably similar, while ensuring that

the final test set is never used for model or threshold selection.

To feed the data into the autoencoder, we use the window length = 30 and stride = 1. For each starting index t, we take the multivariate segment:

$$[x_t, x_{t+1}, \dots, x_{t+29}]$$

and flatten it into a single vector in $\mathbb{R}^{30\times10}$. Each such 30-step window becomes one input sample to the AE with Gaussian noise, allowing the network to implicitly learn short-term temporal patterns within the window even though the model itself is feed-forward. The window label is defined as anomalous if any time point inside the window is labeled as anomaly. Finally, we apply feature standardization to all sensor dimensions. This step removes scale differences between sensors, stabilizes the training of the deep autoencoder, and yields a more reasonable covariance structure for the subsequent MCD-based Mahalanobis distance in the latent space.

3. Methodology

This project includes Autoencoder with Gaussian noise to be robust when training. so we added the noise to our autoencoder

3.1 Autoencoder

(3.1.1) Gaussian noise layer:

We define a custom PyTorch layer that adds Gaussian noise to its input during training. This can help improve model robustness and prevent overfitting.

During training:

$$\tilde{x} = x + \varepsilon \ , \ \varepsilon \sim N(0, \sigma^2 I)$$

where σ is the noise level (set to $\sigma = 0.03$in the experiments) and I is the identity matrix. At inference time no noise is added, i.e. $\tilde{x} = x$

(3.1.2) Encoder and Decoder:

The encoder $f_\theta$maps the (possibly corrupted) input $\tilde{x}$to a low-dimensional latent vector $z \in \mathbb{R}^k$(with $k = 6$):

$$z = f_\theta(\tilde{x}) \in \mathbb{R}^k.$$

Concretely, the encoder is a feed-forward network

$$\tilde{x} \overset{W_1, b_1}{\rightarrow} h_1 = \text{Dropout}\big(\text{ReLU}(W_1\tilde{x} + b_1)\big) \in \mathbb{R}^{64}, h_2 = \text{Dropout}\big(\text{ReLU}(W_2 h_1 + b_2)\big) \in \mathbb{R}^{32},$$

$$z = W_3 h_2 + b_3 \in \mathbb{R}^6,$$

which matches the Encoder class (layers 300→64→32→6 with ReLU and dropout).

The decoder $g_\phi$maps the latent vector back to the input space:

$$\hat{x} = g_\phi(z) \in \mathbb{R}^d,$$

with the structure

$$g_1 = \text{ReLU}(V_1 z + c_1) \in \mathbb{R}^{32}, g_2 = \text{ReLU}(V_2 g_1 + c_2) \in \mathbb{R}^{64}, \hat{x} = V_3 g_2 + c_3 \in \mathbb{R}^d,$$

which corresponds to the Decoder class (6→32→64→300).

The autoencoder is trained only on normal windows using the mean squared reconstruction error (MSE) as the loss function. Optimization is done with AdamW (learning

rate 10^{-3}, weight decay 10^{-4}) in mini-batches of size 128, up to 60 epochs. A hold-out part of the training windows (10%) is used as a validation set, and early stopping with patience 10 is applied. For each epoch the training and validation losses are recorded and later visualized to verify convergence.

3.2 Latent-Space Modeling with Robust MCD

After the autoencoder is trained, all training windows are passed through the encoder to obtain latent codes $z_t \in \mathbb{R}^6$. "Our monitoring statistic is the (robust) Mahalanobis distance in the autoencoder latent space. This is equivalent to a Hotelling's T^2chart with robust mean/covariance estimated by MCD but applied to learned features rather than raw variables; control limits are set empirically by training quantiles. To obtain a robust description of the "normal" latent distribution, we fit a Minimum Covariance Determinant (MCD) estimator on the latent codes of all training windows $\{z_i\}_{i=1}^{N}$. MCD selects a subset $H$of size $h$whose covariance determinant is minimized, and then computes the mean and covariance from that subset:

$$H^{\backslash *} = \arg\min_{|H|=h} \det(\Sigma(H)), \hat{\mu} = \mu(H^*), \hat{\Sigma} = \Sigma(H^*).$$

Compared to the usual sample mean and covariance, this robust estimate $(\hat{\mu}, \hat{\Sigma})$is much less affected by occasional outliers in the training data. Based on this robust Gaussian model, the (squared) Mahalanobis distance of a latent vector $z$is defined as

$$d^2(z) = (z - \hat{\mu})^{\top} \hat{\Sigma}^{-1} (z - \hat{\mu}), d(z) = \sqrt{d^2(z)}.$$

The scalar $d(z)$is used as the primary statistic for the Shewhart chart. For interpretation, we further decompose the distance into per-dimension contributions. Let the inverse covariance

admit a Cholesky factorization

$$\hat{\Sigma}^{-1} = LL^{\top}, u = L(z - \hat{\mu}).$$

Then

$$d^2 = u^{\top}u = \sum_{j=1}^{k} u_j^2 ,$$

so, the term $\text{contrib}_j = u_j^2$ can be viewed as the contribution of the j-th latent dimension to the overall distance. These contributions are later used to explain which latent directions are most responsible for a detected anomaly.

In parallel with the latent distance, the method also keeps the per-window reconstruction error from the autoencoder. For a window x_i with reconstruction $\hat{x}_i$, the mean squared reconstruction error is

$$\text{re}_i = \frac{1}{p}\sum_{j=1}^{p}(x_{ij} - \hat{x}_{ij})^2,$$

and we apply logarithmic transform

$$\tilde{\text{re}}_i = \log(1 + \text{re}_i)$$

to compress very large values and stabilize the scale. Both the latent Mahalanobis distance d_i and the transformed reconstruction error $\tilde{\text{re}}_i$ are treated as anomaly scores derived from the same latent-space model and will be combined with control charts in the next subsection.

3.3 Control-Chart Statistics and Empirical Control Limits

Three scalar statistics are monitored for each window:

- **Latent Mahalanobis distance** d_t(from the MCD model).
- **EWMA-smoothed distance** z_t, computed recursively as

$$z_t = \lambda d_t + (1 - \lambda) z_{t-1},$$

where $\lambda$is the EWMA smoothing parameter. In the base setting, $\lambda = 0.2$, and the initial value is set to the mean of training distances; during scoring, the last EWMA value is carried over as state.

- **Transformed reconstruction error** $\log(1 + \mathrm{RE}_t)$.

Instead of relying on theoretical distributions, all upper control limits (UCLs) are set empirically from training data using quantiles:

- Shewhart UCL for MD:

$$\mathrm{UCL}_{\mathrm{Shewhart}} = Q_q(d_{\mathrm{train}})$$

- EWMA UCL:

$$\mathrm{UCL}_{\mathrm{EWMA}} = Q_q(z_{\mathrm{train}})$$

- Reconstruction-error UCL:

$$\mathrm{UCL}_{\mathrm{RE}} = Q_q(\log(1 + \mathrm{RE}_{\mathrm{train}}))$$

where $Q_q(\cdot)$is the empirical q-quantile of the corresponding training statistics. The base configuration uses $q = 0.997$, which roughly corresponds to a high in-control coverage, and later experiments scan over $q \in \{0.99, 0.995, 0.997, 0.999\}$to see how the false-alarm vs. miss-detection trade-off changes.

For visualization, all three statistics are plotted over time together with their UCLs, and the true anomaly segments are shaded so that the alignment between excursions and labelled

faults can be inspected.

3.4 Decision Logic and Point-Adjusted Evaluation

At the window level, three binary flags are defined for each window:

$$I_t^{(d)} = 1 \quad \text{if } d_t > \text{UCL}_{\text{Shewhart}}$$

$$I_t^{(z)} = 1 \quad \text{if } z_t > \text{UCL}_{\text{EWMA}}$$

$$I_t^{(\text{RE})} = 1 \quad \text{if } \log(1 + \text{RE}_t) > \text{UCL}_{\text{RE}}$$

These flags are combined by three simple logical rules:

- **OR:** anomaly if any of the three flags is 1 (aggressive, high recall).
- **AND:** anomaly only if all (or both, when RE is disabled) flags are 1 (conservative, high precision).
- **VOTE:** anomaly if at least two out of three flags are 1 (compromise).

For evaluation, window-level predictions are first mapped back to point-level decisions. A sliding-window aggregation function assigns to each time index the maximum predicted value over all windows covering that index. To better reflect practical requirements on SMD, a point-adjusted (PA) version of the labels is also computed: if any point inside a true anomalous segment is predicted as anomalous, the whole segment is marked as detected. This raises recall on long segments and avoids over-penalizing small temporal shifts.

Standard classification metrics—accuracy, precision, recall, F1-score—are computed both on the raw point-wise predictions and on the PA labels. In addition, a fused continuous score is defined by normalizing each statistic with its UCL and taking the maximum

$$s_t = \frac{d_t}{\text{UCL}_{\text{Shewhart}}}, e_t = \frac{z_t}{\text{UCL}_{\text{EWMA}}}, r_t = \frac{\log(1 + \text{RE}_t)}{\text{UCL}_{\text{RE}}}, f_t = \max(s_t, e_t, r_t),$$

which is used for ROC-AUC and PR-AUC as well as for histograms of normal vs. anomalous windows.

4. Experiments and Results

4.1 Baseline setup

We use all models in Pytorch and the python package minimum covariance determinant (MinCovDet) to estimate the robust mean and covariance in latent space. The environment of Python is 3.12.9 and we set seed at 42. We set the Gaussian noise with standard deviation at 0.03 to improve robustness and reduce overfitting. The model is trained with batch size 128 for up to 60 epochs, using AdamW with learning rate $1 \times 10^{-3}$and weight decay 1×10^{-4}. Early stopping is applied with patience = 10 based on validation loss. **Figure 1** shows the training/validation loss curves. Both curves decrease quickly in the early epoch and then gradually flatten, indicating that the model successfully learns a stable representation of normal patterns.

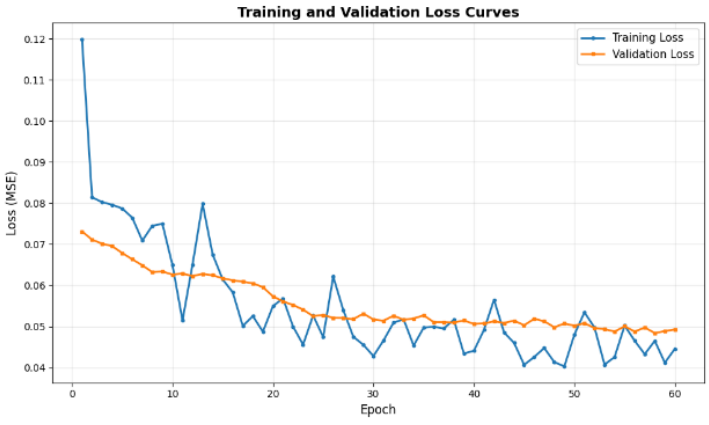


Figure 1. Training and Validation Loss Curves

4.2 Baseline Performance on Validation Set

This section we evaluate the proposed monitoring pipeline on the validation split using a fixed baseline configuration. The purpose is to establish a clear reference point before conducting sensitivity analyses and hyperparameter tuning. The baseline setting uses an empirical control-limit quantile of $q = 0.997$, EWMA smoothing parameter $\lambda = 0.2$, and a decision logic of OR. **Figure 2** shows that control chart that monitoring charts applied to learned anomaly statistics. Specifically, we monitor (i) the Mahalanobis distance in latent space with a Shewhart-style upper control limit (UCL), (ii) an EWMA-smoothed version of the score with smoothing parameter λ, and (iii) the reconstruction error from the autoencoder. The UCLs are empirically calibrated using the distribution of normal training data (quantile-based limits), enabling threshold-based statistical process monitoring of anomaly scores Outside the shaded anomaly segments, most of the statistics remain below the UCLs, indicating stable in-control behavior. However, occasional isolated exceedances may still appear in long normal regions, which correspond to false alarms under the OR decision rule. **Table 1** shows that report standard point-wise metrics computed at the window level. These metrics quantify classification performance under the strict requirement that each individual window must be labeled correctly. In anomaly detection, point-wise metrics are often dominated by the large number of normal windows (class imbalance), so they may under-represent the practical value of early or partial detection within an anomaly segment. To better reflect event-level detection quality, we also report point-adjusted, which aligns with operational requirements where detecting the occurrence of a fault is more important than labeling every timestamp precisely.

Type	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Point-wise	0.849	0.300	0.986	0.460	0.948	0.684

Point adjust	0.850	0.303	1.000	0.465	0.920	-----

Table 1 Point-wise and point adjust metrics on Validation

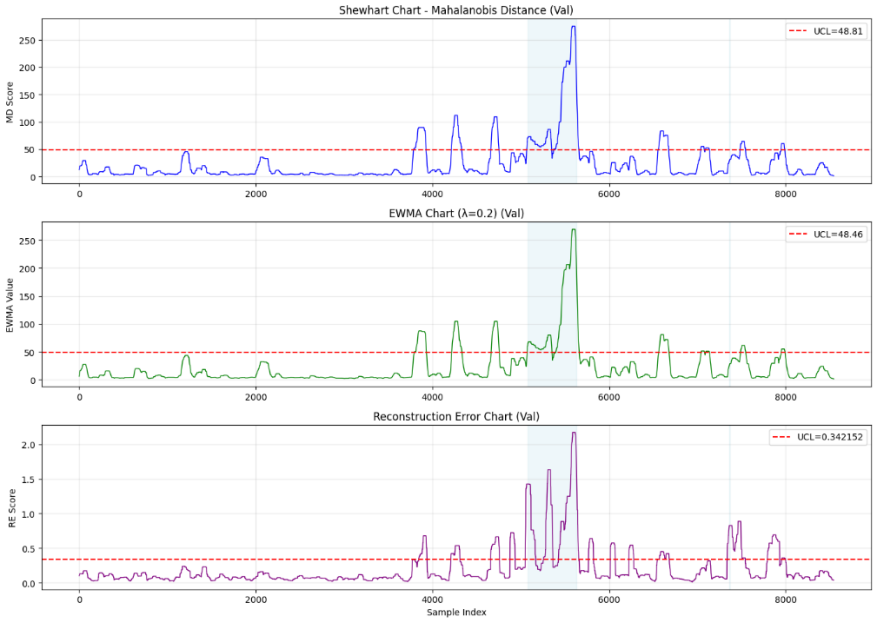


Figure 2 Shewhart MD, EWMA, Reconstruction Error on val.

4.3 Sensitivity Analysis: Effect of q (UCL Quantile) and Effect of Decision Logic (OR / VOTE / AND)

We set the λ=0.2, logic=OR and q∈{0.99,0.995,0.997,0.999} This bar chart evaluates how the choice of the upper control limit (UCL) quantile q affects point-adjusted (PA) Precision, Recall, and F1-score on the validation set. Increasing q makes the UCL more conservative (higher threshold), which typically reduces the number of triggered alarms. In our results, PA-Recall remains at 1.0 for all tested q, indicating that every anomaly segment is still detected at least once regardless of the threshold. Meanwhile, PA-Precision increases monotonically as q increases (from ~0.27 at q=0.990q=0.990 to ~0.37 at q=0.999q=0.999), suggesting fewer false alarms on normal segments. As a result, PA-F1 also improves with larger q, reflecting a better precision–recall balance under event-level (segment-based)

evaluation. Overall, this analysis shows that using a higher quantile (e.g., q=0.997, q=0.997–0.999) can significantly reduce false positives while preserving near-perfect event detection in PA evaluation.

To combine multiple detection signals such as Shewhart/MD, EWMA, and reconstruction error, we compared three fusion logics on the validation set under point-adjusted (PA) evaluation. As the fusion rule becomes stricter (OR → VOTE → AND), precision increases while PA-recall stays nearly unchanged and AND gives the best trade-off on VAL.

Fusion logic	Decision rule	PA Precision	PA Recall	PA F1
OR	Alarm if any detector triggers	0.303	1.000	0.465
VOTE	Alarm if majority triggers	0.450	0.995	0.619
AND	Alarm if all detectors' triggers	0.535	0.995	0.696

Table 2 Effect of Fusion Logic (OR vs. VOTE vs. AND)

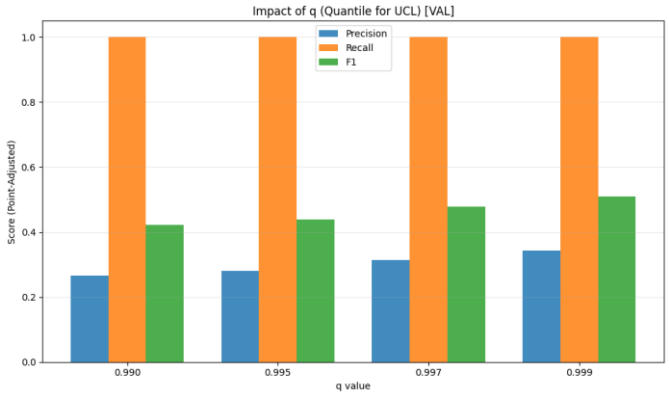


Figure 3. Impact of the UCL quantile q on point-adjusted detection performance (VAL)

4.4 Hyperparameter Grid Search on Validation Set

To improve the detection, we further conduct the grid search on the validation set over

three key hyperparameters: the empirical UCL quantile q, the EWMA smoothing parameter λ, and the decision fusion logic.

$$q \in \{0.99, 0.995, 0.997, 0.999\}\ ,\ \lambda \in \{0.1, 0.2, 0.3\},\ \text{logic} \in \{\text{OR}, \text{VOTE}, \text{AND}\}$$

For each, we keep the trained autoencoder and the fitted MCD model fixed and only vary the charting thresholds and the alarm decision rule. We then compute PA-F1, which reflects the overall segment-level detection quality under the class imbalance setting. From Table 3, We can compare with baseline setting, the best hyperparameter is q = 0.999 with λ = 0.1, logic is AND, provides a strong event-level trade-off: it detects anomaly segments reliably without being overwhelmed by false alarms.

Rank	q	λ	logic	PA-precision	PA-recall	PA-F1(from score)
1	0.999	0.1	AND	0.982	0.994	**0.988**
2	0.999	0.3	AND	0.897	0.994	0.943
3	0.999	0.1	VOTE	0.788	0.994	0.879
4	0.995	0.1	VOTE	0.771	0.994	0.869
5	0.999	0.2	AND	0.748	0.994	0.854

Table 3 Grid Search Results Top 5

4.5 Final Test Results on Held-out Final Set and Sanity Check on Full Test Sequence

We report the detection performance on a held-out Final split that is not used for threshold selection or hyperparameter ranking. **Table4** shows the best configuration improves PA-F1 on the Final relative to the line. The gain mainly comes from (higher precision, higher recall, a balanced change in both indicating that fewer false alarms and fewer missed anomaly segments are achieved after tuning the control limits and fusion logic. **Figure 4** shows the Shewhart chart on latent Mahalanobis distance and EWMA chart on the

MD statistic and reconstruction error chart. Overall, the three charts provide complementary evidence—MD (Shewhart) detects abrupt deviations, EWMA emphasizes persistent shifts, and reconstruction error captures manifold mismatch—while the shaded anomaly regions show consistent UCL exceedances that align well with labeled faults on the FINAL test set. **Figure 5** shows the final fused anomaly score on the held-out final test set, separated by Normal vs. Anomaly windows. The normal windows are heavily concentrated near low scores (close to 0), indicating that most in-control behavior is assigned to low anomaly evidence. In contrast, anomaly windows exhibit a right-shifted distribution with more mass at larger scores, showing that abnormal conditions are generally assigned higher anomaly evidence. **Figure 6** shows the sanity check as an additional robustness check, we run full test set sequence. It increases confidence that the proposed AE + robust MCD latent modeling combined with Shewhart/EWMA monitoring behaves reliably beyond a short evaluation window, and that the empirical control limits remain effective over time.

Setting	split	q	λ	logic	PA-precision	PA-recall	PA-F1
Baseline	Final	0.997	0.2	OR	0.507	0.999	0.673
Best	Final	0.999	0.1	AND	0.978	0.998	0.988

Table 4 Best vs. Baseline

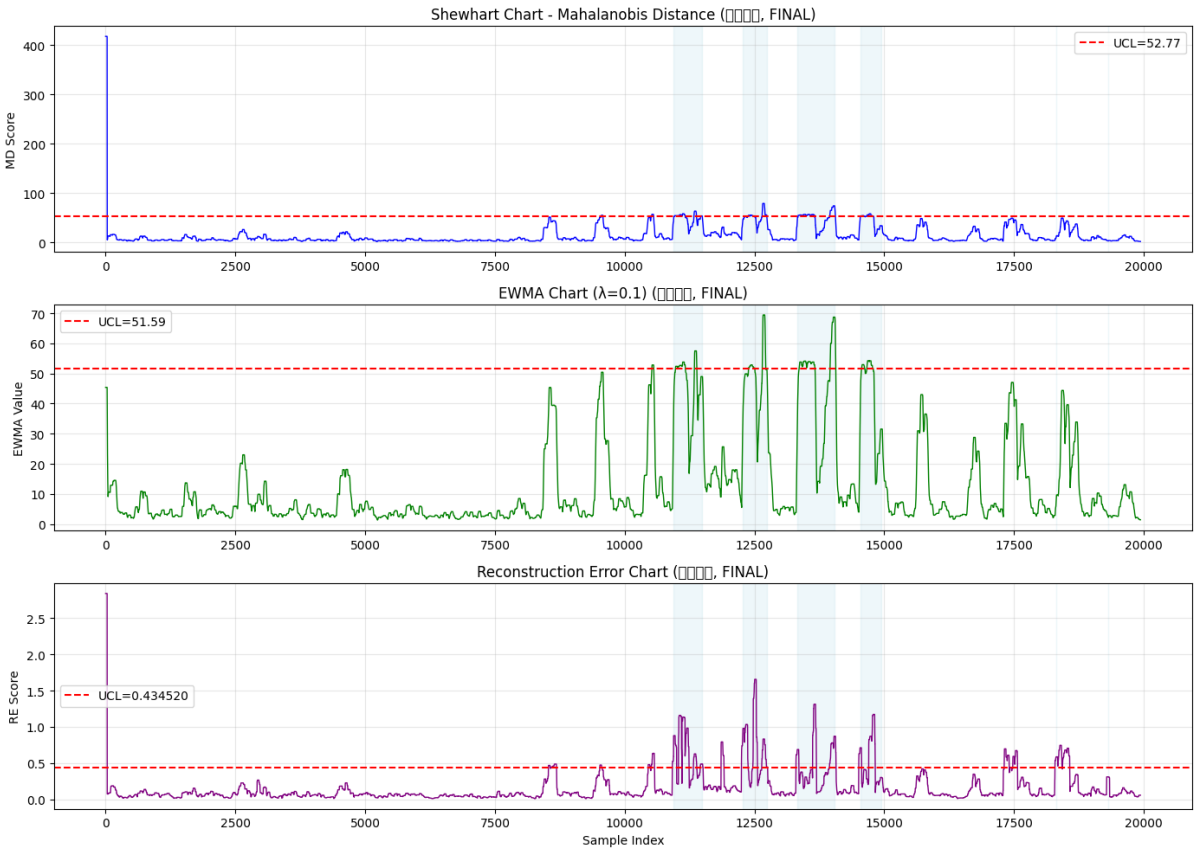


Figure 4 Control charts on FINAL set (best hyperparameters)

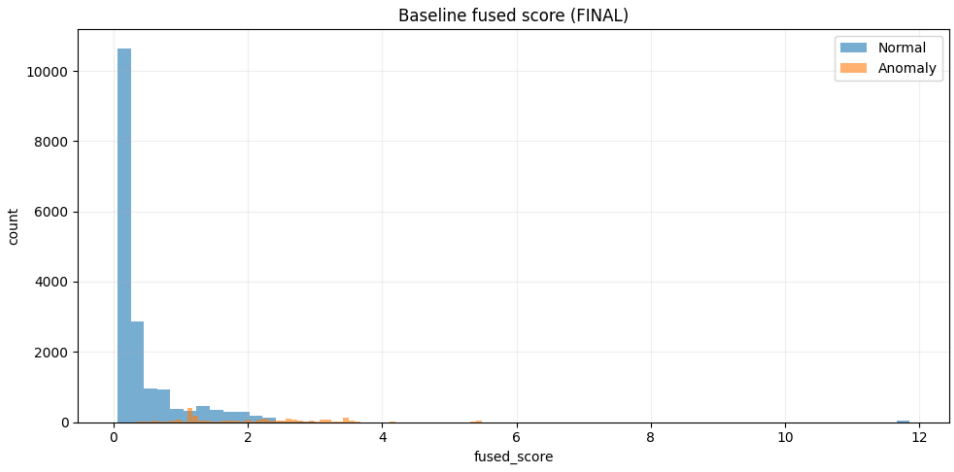


Figure 5 Final fused anomaly score



Figure 6 Sanity check

4.6 Case Study

We also construct the Reconstruction Example. **Figure 7** compares the original input signals (solid line) and the autoencoder reconstructions (dashed line) for several representative samples. Across samples, the reconstruction captures the dominant periodic patterns and overall waveform structure reasonably well, indicating that the model has learned the typical “normal” dynamics. However, noticeable discrepancies occur around sharp spikes and abrupt changes, where the reconstructed signal smooths or underestimates the peak magnitude. This behavior is expected in reconstruction-based anomaly detection: patterns that deviate from the learned normal manifold are harder to reconstruct accurately, leading to a larger reconstruction error (RE) and higher anomaly scores. The examples illustrate how the autoencoder effectively models regular patterns while struggling with anomalous or rare events, which is the basis for using reconstruction error as an anomaly detection signal. **Figure 8** shows the traditional SPC chart. The PCA-MSPC, which contains

the Hotelling T^2 Control Chart and Q (SPE) Control Chart in our dataset. $\mathbf{T^2}$ is typically sensitive to changes such as mean shifts or global pattern drifts along dominant normal directions. Q (SPE) is therefore sensitive to structural deviations that cannot be captured by the normal low-dimensional manifold.

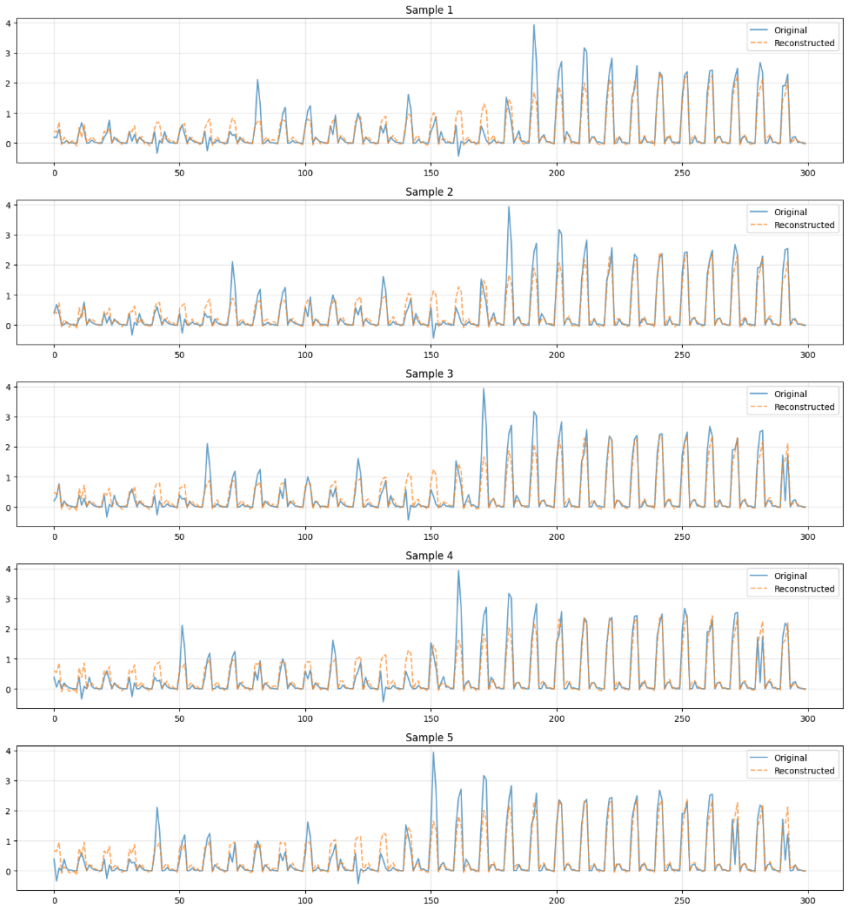


Figure 7 Reconstruction Examples (Original vs. Reconstructed Signals)

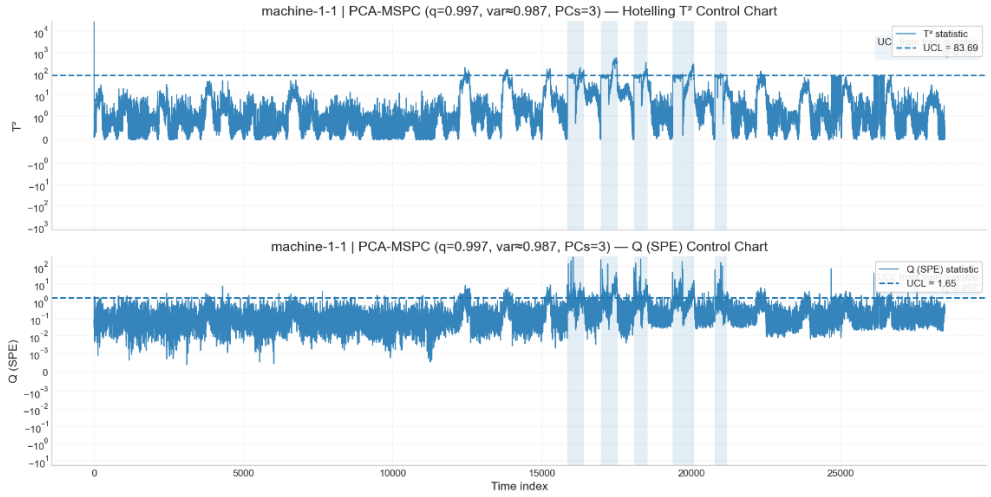


Figure 8 PCA-MSPC control charts on the test sequence

5. Discussion

The proposed detector monitors three complementary statistics: latent-space Mahalanobis distance (MD), EWMA-smoothed MD, and reconstruction error (RE). In practice, we observe that these three signals often trigger in different ways across anomaly segments, which supports the motivation for fusing them: the ensemble is more robust than any single statistic. The detection performance depends strongly on the control-limit quantile q, EWMA parameter λ, and the fusion logic. OR is aggressive (high recall), AND is conservative (higher precision), and VOTE often provides a balanced default. Based on our validation study, we recommend VOTE (or OR if missing faults is expensive) as a practical default, and AND as a conservative mode when false alarms are costly. A key benefit of using MCD is that the estimated mean/covariance are less influenced by a small fraction of outliers in the training data, leading to more stable MD scores and more reliable empirical control limits. The limitation is that this project only on SMD machine1-1 with ten sensors. so cross-machine generalization is not validated

6. Conclusion

We use a hybrid method to anomaly detection framework that combines a autoencoder with Gaussian noise for representation learning with robust latent-space modeling via MCD. After training, windows are mapped into a low-dimensional latent space, where a robust Gaussian model produces a Mahalanobis-distance statistic. Together with EWMA smoothing and reconstruction error, the method forms a three-statistic monitoring system with empirically estimated control limits. Selected from the SMD dataset machine 1-1 of 10 sensors, the framework achieves strong point-adjusted performance under the best configuration.

7. References

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Code:

The source code is on **GitHub repository**..

Click the GitHub repository